



RIVER VALLEY HIGH SCHOOL

JC 2 PRELIMINARY EXAMINATION

CANDIDATE
NAME

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CENTRE
NUMBER

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CLASS

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INDEX
NUMBER

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H2 BIOLOGY

9744/04

Paper 4 Practical

17 Aug 2021

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number, class and name on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	

Answer **all** questions.

- 1** In this experiment, you will investigate the effect of substrate concentration on the rate of respiration. Liver tissue was removed from the organism and allowed to respire overnight. Intact mitochondria were then extracted from the liver tissue and placed in a cold buffer solution, labelled **V**.

You are required to:

- make initial observations to determine how to measure the rate of respiration
- perform serial dilution to obtain different concentrations of substrate **S**
- collect results and present it in a suitable table

You are provided with:

- 1.0 % substrate, in a beaker labelled **S**
- mitochondria suspension, in a beaker labelled **V**
- distilled water, in a beaker labelled **W**
- DCPIP solution, in a container labelled **DCPIP**

DCPIP is an irritant. Suitable eye protection should be worn. If DCPIP comes into contact with your skin, wash it off immediately under tap water.

Before starting the investigation, read through steps 1-13 and prepare a table in (c).

Proceed as follows.

- 1** Using the scissors, cut at least 10 squares of filter paper of dimension 1 cm by 1 cm.
- 2** Prepare mixture **D** by mixing suspension **V** and solution **DCPIP** in a ratio of 15 to 1. Prepare at least 20 cm³ of mixture **D** for step 3.
- 3** Pour mixture **D** into the petri dish to fill it to a height of approximately 2 mm.
- 4** Place the filter papers into the petri dish. Make sure that the filter papers are completely covered by mixture **D**.
- 5** You are required to make a serial dilution of 1.0 % substrate **S**, to reduce its concentration by **half** between each successive dilutions.

After the serial dilution is completed, you will need to have at least 40 cm³ of each concentration available for use.

(a) Complete Fig. 1.1 to show how you will dilute **S**.

[3]

For each beaker:

- state, under the beaker, the volume and concentration of the substrate **S** in the beaker that will be available for use in the investigation, after the serial dilution has been completed
- use one arrow, with a label above the beaker, to show the volume and concentration of **S** added to prepare the solution in the beaker
- use another arrow, with a label above the beaker, to show the volume of distilled water, **W**, added to prepare the concentration of **S** in the beaker.

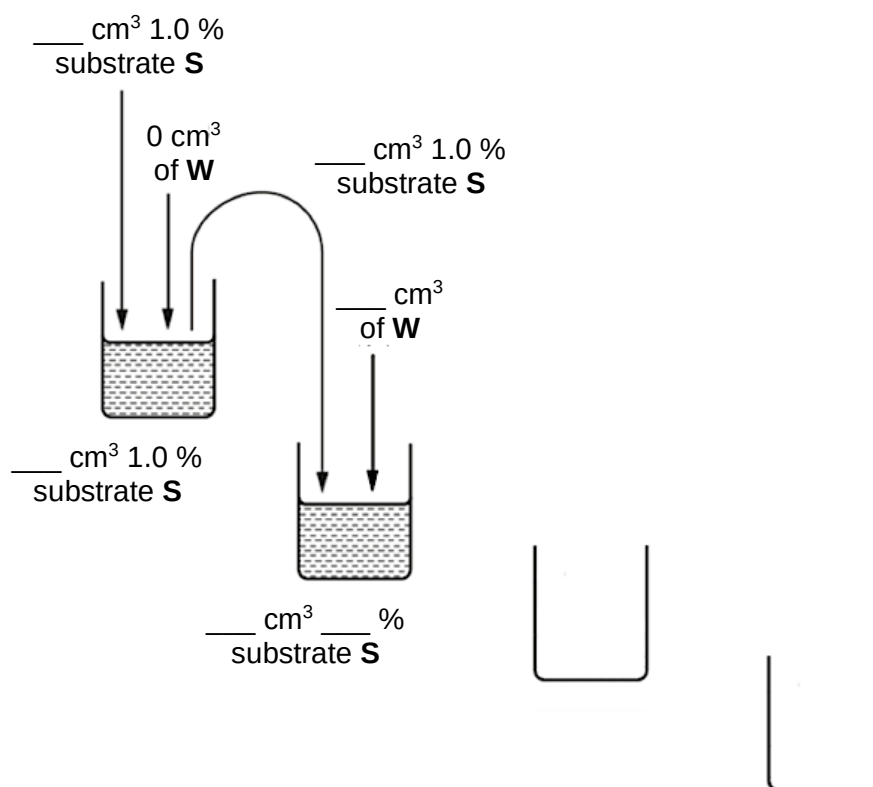


Fig. 1.1

- Using a marker, draw a line to indicate on the specimen tube the position 3 cm from the top of the tube.
 - Pour substrate **S** of concentration 1.0 % into the specimen tube up to this mark.
 - Use the forceps to pick up one square of filter paper from the petri dish. Wipe the square against the petri dish to remove excess solution from both sides of the square.
 - Place the square at the bottom of the specimen tube. Immediately release the square and observe it for one minute.
- (b) Describe your observations on the appearance and behaviour of the square of filter paper. [2]

10 Remove the square from the specimen tube and empty the contents of the specimen tube into the container labelled **for waste**.

11 Using your observations in **(b)**, you are required to decide on what you will measure to determine the rate of respiration. You should choose a dependent variable that allows you to observe a trend in the concentration range of substrate **S** prepared.

Describe the dependent variable.

12 Use the method you have decided to collect results for all concentrations of **S**. If the time taken is more than 1 minute, record the results as 'more than 60.0'.

13 Calculate the rate of respiration and record in the table in **(c)**.

(c) Record your results in a suitable table in the space below.

[4]

(d) (i) Complete Table 1.1 to:

- identify one significant source of error for your method

[2]

- suggest how to make an improvement to increase the accuracy of your results in **(c)**.

Table 1.1

significant source of error	how to make an improvement

- (ii)** Suggest the identity of substrate, **S**.

Give a reason for your answer.

[2]

identity _____

reason _____

- (e)** Another experiment was conducted to investigate the rate of respiration in different species of organisms, **X**, **Y** and **Z**. Intact cells were obtained from the same tissue of the different organisms and placed separately in cold buffer solutions.

In the experiment, all other conditions, including the amount of oxygen (in excess), the amount of glucose and the number of cells, were kept the same. 20 minutes after the start of the experiment, a small volume of reaction mixture was removed

and Benedict's test was conducted to measure the time taken for first colour change.

- (i) Explain how the rate of respiration can be shown by the time taken for first colour change. [2]

Further investigation was made using the same experimental setup to determine the effect of the amount of protein present in the inner mitochondrial membrane on the time taken for first colour change in the Benedict's test. Table 1.2 shows this relationship.

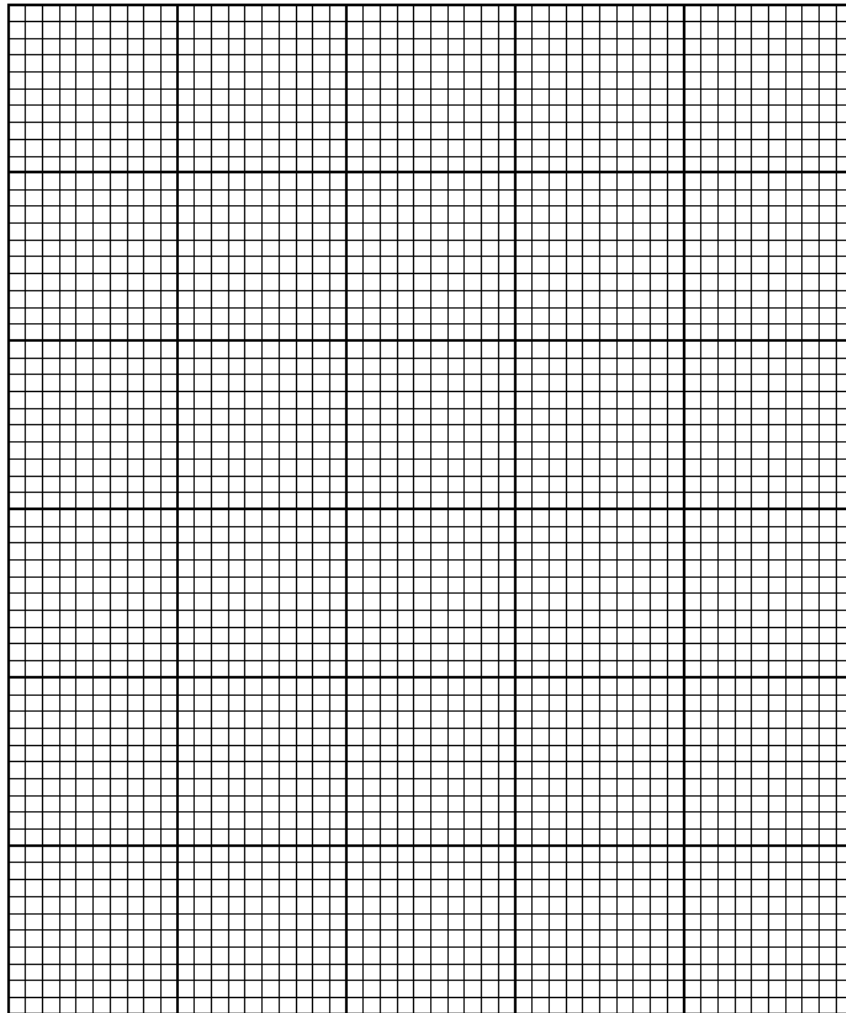
Using spectroscopic techniques, absorbance measured at 280 nm (A_{280}) was used to determine the amount of protein present in the **inner mitochondrial membrane**.

The higher the absorbance value, the more proteins present.

Table. 1.2

A_{280} / a.u.	time taken for first colour change / s
1.25	280
1.00	202
0.75	126
0.50	78
0.25	65

- (ii) On the grid provided, plot a graph of the data shown in Table 1.2. [3]



(iii) Using **(e)(ii)**, explain what is meant by “limiting factor”.

[2]

(iv) In the same investigation, during anaerobic conditions, it was found that when A_{280} was 0.75 a.u., the time taken for first colour change was 166 s.

Using the information provided and the same absorbance range, draw the predicted graph for anaerobic conditions on the grid in **(e)(ii)**.

[1]

- (v) Explain the shape of the graph for (e)(iv) and the difference in time taken for first colour change between aerobic and anaerobic conditions when A_{280} was 0.75 a.u.. [2]

The results for the experiment on X, Y and Z are shown in Table 1.3.

Table 1.3

organism	time taken for first colour change / s
X	135
Y	224
Z	92

- (vi) The absorbance at 280 nm (A_{280}) can be used to estimate the amount of protein in a mixture using the following formula:

$$\text{concentration (mg / ml)} = A_{280} \times \text{path length of spectrometer (cm)}$$

Given that the path length is 0.8 cm and there are 250 000 cells in 1.5 ml of the reaction mixture tested, use the graph and calculate the amount of protein per cell for the organism with the highest rate of respiration.

Show **all** working clearly. [2]

[Total: 25]

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- 2 Concentrated biological washing liquids are alkali solutions that contain enzymes and other chemicals to help remove stains caused by biological molecules. One of these enzymes is lipase. A student found a website that lists the concentrations of lipase in the concentrated biological washing liquids. These concentrations vary between 85 mg cm^{-3} and 155 mg cm^{-3} . These concentrated biological washing liquids need to be diluted with water to reach the working concentration of enzyme lipase.

The recommended dilutions for washing fabrics using these concentrated biological washing liquids are:

- Washing liquid **A** – 20 cm³ of washing liquid in 5000 cm³ of water
- Washing liquid **B** – 35 cm³ of washing liquid in 10 000 cm³ of water.

The student was asked to find the working concentration of enzyme lipase for biological washing liquids, **A** and **B** when diluted.

- (a) The student used the information from the website and the recommended dilutions to calculate the working concentration of lipase that could be present in diluted biological washing liquids **A** and **B**.

The student first calculated the **maximum** possible working concentration of lipase in washing liquid **A** when diluted. The method is shown below.

The maximum possible concentration of lipase in concentrated washing liquid **A** is 155 mg cm⁻³. Therefore, the maximum possible working concentration of lipase in diluted washing liquid **A** is:

$$\frac{(155 \text{ mg cm}^{-3} \times 20 \text{ cm}^3)}{5000 \text{ cm}^3} = 0.62 \text{ mg cm}^{-3}$$

Calculate the **minimum** possible concentration of lipase in diluted washing liquid **B**.

Show all working in the space provided.

[2]

minimum possible concentration of lipase in diluted washing liquid **B**: _____

To find out the exact working concentrations of lipase in diluted washing liquid **A** and **B**, the student made five different concentrations of lipase, 0.2%, 0.4%, 0.6%, 0.8% and 1.0%, and simulated fat-stained fabric by soaking them in cooking oil.

He diluted the concentrated washing liquids **A** and **B** according to the recommended dilutions. The website also recommends that the washes using the diluted washing liquids should be conducted for 1 hour at 30 °C, instead of the optimum temperature of lipase, which is 60 °C.

As the reaction between lipase and cooking oil on the fat-stained fabric proceeds, the pH of the mixture will decrease because of the fatty acids produced from the hydrolysis reaction.

- (b) Explain how a lower temperature of 30 °C affects the results of the experiment. [2]

- (c) Suggest why the recommended temperature to use for washing is 30 °C instead of lipase's optimum temperature of 60 °C. [1]

- (d) Using apparatus commonly found in a laboratory and the materials below, outline a method the student could use to find the working concentration of lipase in the diluted washing liquid **A** and **B** with high accuracy and reliability.

Data table and risk assessments are **not** needed.

- 0.2%, 0.4%, 0.6%, 0.8% and 1.0% lipase solutions
- diluted washing liquid **A**
- diluted washing liquid **B**
- 10 cm x 10 cm fat-stained fabric

[8]

- strings
- boiling tubes
- scissors
- 1% sodium hydroxide
- bromothymol blue which is blue at alkali pH and turns yellow at pH 6.0
- stopwatch

[illegible]

- (e) The student conducted the experiment and his results suggested that diluted washing liquid **A** had a higher working concentration of lipase than diluted washing liquid **B**.

Suggest one reason why this conclusion may **not** be valid.

[1]

[Total: 14]

- 3 During this question you will require access to a microscope.

You are provided with:

- a piece of eggplant fruit, **B1**
- iodine solution, in a container labelled **iodine**

Iodine is a stain. If iodine comes into contact with your skin, wash it off with cold water.

You are required to:

- record observations following the addition of **iodine**
- make a microscope slide of this sample
- draw cells in **B1** observed at high magnification
- calculate the actual size of one cell in **B1**.

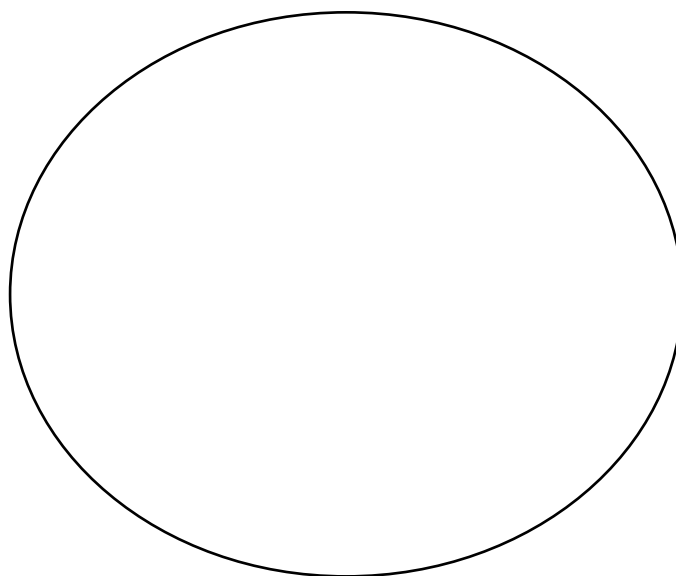
Proceed as follows.

1. Cut two transverse slices of eggplant, **B1**. Each slice should be approximately 3 mm thick. Do not remove the skin.
2. Put 5 to 8 drops of **iodine** into one Petri dish.
3. Put one slice of eggplant into the Petri dish. Add more drops of **iodine** to cover the top surface of the slice.
4. Cover the Petri dish with its lid and leave the slice of eggplant for one minute.
5. Observe the slice of eggplant stained with **iodine** through the lid of the Petri dish.

- (a) The diagram provided in the space below shows the outline of the slice of eggplant. On the diagram, draw the stained regions observed in **step 5**.

Add labelling lines and the label 'starch' to identify the regions that has starch present.

[1]



6. Use the forceps to pull out a thin piece of the eggplant from one of the stained regions.
7. Put a drop of water on a glass slide and place the piece of eggplant on the glass slide.
8. Carefully place the cover slip over the piece of eggplant. Place a paper towel on top of the glass slide and press firmly on the cover slip.
9. Use your microscope, at high magnification, view the slide to find two whole cells.

- (b) In the space provided, draw **two** adjacent whole cells that you have selected.

You should include in your drawings any internal structures that you can see.

[3]

Labels are **not** required.

- (c) Using the stage micrometer and the eyepiece graticule fitted in the microscope, calculate the actual size one cell.

Show **all** working clearly.

[3]

In the tomato fruit, one method to determine its ripeness is to measure its pH. Tomatoes become less acidic as they ripen. For a perfectly ripe tomato, the pH of its pulp extract is approximately 4.6.

Indicator **G**, can be used to determine the pH. Indicator **G** is orange-red at acidic pH, green at neutral pH and purple-blue at alkaline pH.

P1, **P2** and **P3** are purified tomato pulp extracts from different tomatoes of the same tomato variety but harvested at different times.

You are required to:

- record observations for **P1**, **P2** and **P3** when tested with indicator **G**

[2]

- determine and rank the tomato pulp extracts according to the level of ripeness.

Proceed as follows:

1. Add one drop of **G** to one drop of **P1**.
2. Record observations in Table 3.1.
3. Repeat steps 1-2 using **P2** and **P3**.
4. Rank the tomato pulp extract based on its ripeness where 1 indicates unripe and 3 being the most ripe.

(d)

Table 3.1

	P1	P2	P3
observations			
ripeness			

Turn over for the remainder of Question 3

Fig. 3.1 is a photomicrograph of the stem cross-section of a tomato plant while Fig. 3.2 is a photomicrograph of the stem cross-section of another species of plant.

You are not expected to be familiar with this specimen.

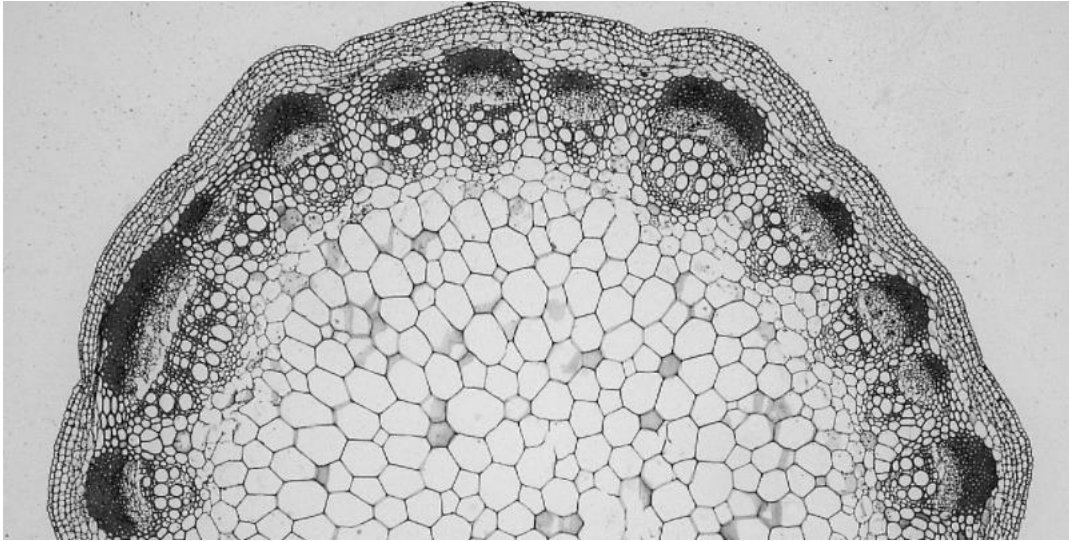


Fig. 3.1

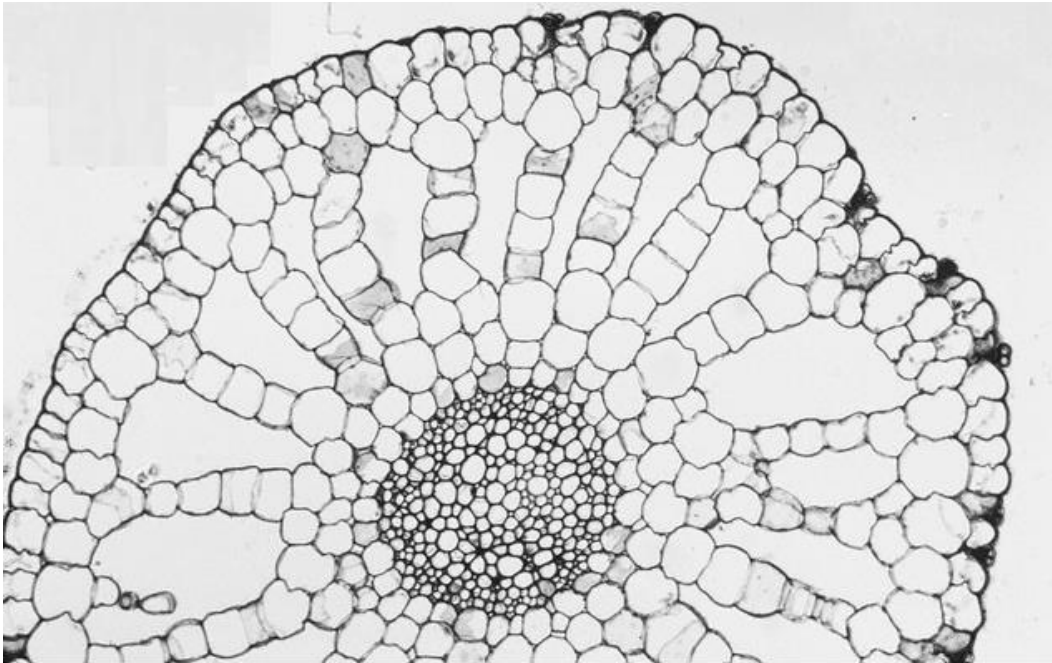
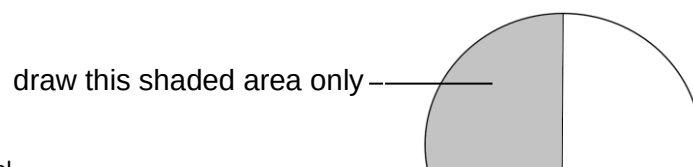


Fig. 3.2

- (e) Use the space provided to draw a large plan diagram of the part of the stem shown in Fig. 3.2.



[4]

A plan diagram shows the arrangement of different tissues, including their correct shape and proportions. No cells should be drawn.

Labels are **not** required.

- (f) Using a suitable form, distinguish between the stem structures shown in Fig 3.1 and Fig 3.2. [3]

[Total: 16]

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